Basic Electrical and Electronics Engineering

Semester	Course Code	Name of the course	L	T/A	P	Credits
B.Tech. First Year /Sem I / Sem-II		Basic Electrical and Electronics Engineering	2	0	0	2

Prerequisites for the course	
1	Basic of Electrical and Electronics Engineering such as PN junction diode, electrons, flow of current transistor etc.

Prior Reading Material/useful links	
1	Basic Electrical and Electronics Engineering by S.K. Bhattacharya
2	https://www.youtube.com/watch?v=YEgWrhxZjlw
3	https://www.youtube.com/watch?v=q2L5s7i4FrI&list=PL0s3O6GgLL5cLAfoALo36QVhy1oM5NZsP&index=2

Course Outcomes:

Sr. No.	Course outcome number	CO statement
1	CO 1	Define fundamentals of electrical system and choose measuring instruments for measurement of electrical quantities & describe the concept PN junction diode and its characteristics.
2	CO 2	Classify wiring system and compare energy resources for electrical energy generation & elaborate the transistor configuration in CE, CB & CC mode.
3	CO 3	Plan and organize the utilization of energy resources of electrical system & apply transistor characteristics to construct Amplifier devices.
4	CO 4	Compare different sources of electrical system & distinguish various logic gates and simplify the Boolean's equations.
5	CO 5	Justify the utilization of various electrical and electronics components into electrical and electronics circuitries.
6	CO 6	Construct various circuits using Resistors, capacitors, inductors, PN junction diode, Zener diode, transformers, transistors and logic gates

Syllabus:

	Course Contents
Unit I	Fundamental of Electrical system: Potential difference, Ohm's law, Effect of temperature on resister, resistance temperature coefficient, Electrical wiring system: Study of different wire gauges and their applications in domestic and industry. Resistors: color code, type of resistors, material used for resistors, resistance wires, resistance standards, frequency errors in resistors. Capacitors: Capacitance standards, variable capacitors, frequency errors in capacitors. Loss angle and power factor of capacitors. Inductors: standards of inductance, mutual inductance, self-inductance, variable inductance, inductors for high and low frequency work, frequency errors in inductors.

V.K. Mehta
Basic electrical system

	[6 Hrs]
Unit II	Measurement Of Electrical Quantities, Measuring Instruments & Energy Resources Measurement of Voltage, Current, and Power (1ph and 3ph), Introduction to PMMC instrument, Ohmmeter, galvanometer, potentiometers, power factor meter and frequency meters. Study of circuit breakers & Actuators (MCB & Fuse, Power Contactors & Aux contactors, Electro-Mechanical & Solid state Relays). Energy Resources and Utilization: Conventional and nonconventional energy resources; Introduction to electrical energy generation from different resources, transmission, distribution and utilization, Concept of Supply Demand, Power Factor, Need of unity factor. **[6 Hrs]**
Unit III	Introduction to diodes, diode circuit and Transducers The P-N Junction Diode, V-I characteristics, Diode as Rectifier, specifications of Rectifier Diodes, Half Wave, Full wave, Bridge rectifiers, Equations for IDC VDC VRMS, IRMS, Efficiency and Ripple Factor for each configuration. Zener Diode, Characteristics, Specifications, Zener Voltage Regulator, Types of Diodes: LED, Photodiode. Introduction to transducer, Classification of transducers, characteristics and choice of transducers. **[6 Hrs]**
Unit IV	Semiconductor Devices and Applications: Transistors: Introduction, Classification, CE, CB, and CC configurations, α, β, concept of gain and bandwidth. Operation of BJT in cut-off, saturation and active regions (DC analysis). BJT as an amplifier, biasing techniques of BJT, BJT as a switch. Introduction to Digital Electronics: Number System, Basic logic Gates, Universal Gates, Boolean Postulates, De-Morgan Theorems **[6 Hrs]**
Unit V	Introduction to power system : Importance of power system, main parts of power system, single line diagram, types of transmission (primary transmission and secondary transmission), types of distribution system, substation and its types. **[6 Hrs]**
	Text Books
1	Anurag Kandya, "Elements of Electrical Engineering", Charotar Publishing, Anand
2	M. S. Palani Gamy, "Basic Electronics Engineering", Tata Mc-Graw Hill Publication
	Reference Books
1	K. Venugopal and V. Prabhu Raja, Basics of Engineering, New Age International (P) Ltd, 2008.
2	P. K. Nag "Engineering Thermodynamics", Tata McGraw Hill, New Delhi 3rd ed. 2005
	Useful links
1	https://nptel.ac.in/courses/11210409/1
2	https://www.digimat.in/nptel/courses/video/105104101/L01.html
3	https://www.youtube.com/watch?v=cErccoHui9g&pbjreload=10

A.K. Sawni

Badri Ram
Switch Gear Protecti

V.K. Mehta
Basics of Electronics systems